Case 2 — Mr. Yusuf Ibrahim

62 M | 10 months post-stroke — Left Cerebellar Infarct | Community Physiotherapy Clinic | Cerebellar ataxia Dysmetria (past-pointing) Positive Romberg's Wide-based gait + trunk sway Near-miss falls ×2 Strength intact

Neurological Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Wide-based ataxic gait with staggering steps and trunk sway
- Impaired dynamic balance — especially turns and uneven surfaces
- Visual dependence for balance (Romberg's positive)
- Difficulty in crowds and on uneven terrain
- Reduced independence in outdoor mobility
- Difficulty with quick turns and directional changes

■■ RED FLAGS

- New or suddenly worsening neurological symptoms
- Sudden onset of severe vertigo with vomiting unrelated to head movement:
- May indicate acute vertebrobasilar insufficiency or new posterior fossa event — urgent neurological review.
- Cardiovascular symptoms on exertion
- (chest pain, dyspnoea, syncope, palpitations): Significant vascular risk factor history underlying his stroke — cardiac review and cardiology clearance before exercise intensification.
- Evidence of recent fall with head injury or suspected fracture:
- Medical clearance required before any loading or balance exercises commence.
- Rapidly progressive ataxia:

■ PATHOPHYSIOLOGY (KEY POINTS)

- Ibrahim has sustained an ischaemic infarct of the left cerebellar hemisphere , with probable involvement of the vermis given the prominent truncal ataxia.
- The cerebellar vermis specifically coordinates axial and gait stability, and its damage produces the characteristic wide-based gait, trunk sway, and staggering steps Mr.
- Sign / Symptom Pathophysiological Explanation "Boat rocking" sensation (oscillopsia) Disrupted cerebellar-vestibular integration — the cerebellum normally damps oscillatory vestibular signals.
- Without this damping, a persistent sense of motion is experienced even at rest.
- Dysmetria / past-pointing The cerebellum computes the "brake" signal that terminates a limb movement precisely at the target.
- Without it, movements overshoot or undershoot — visible as finger-nose past-pointing and heel-shin inaccuracy.

■ DIFFERENTIAL DIAGNOSES

1. Why I favour cerebellar ataxia over peripheral vestibular neuritis:

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Key Neurological Observations Before Testing

Tone: Passive movement of left UL and LL — expect ipsilateral hypotonia (cerebellar hypotonia). No spasticity expected; note any that is present as it would suggest additional corticospinal involvement. Coordination: Finger-nose test (dysmetria, intention tremor), heel-shin test (trajectory accuracy), rapid alternating movements (dysdiadochokinesia if present) — all expected to be impaired ipsilaterally. Posture:

Test 1 — Berg Balance Scale (BBS)

What it measures: Static and dynamic balance across 14 functional tasks (sitting, standing, reaching, turning, tandem standing, single-leg stance). Scored 0–56; <45 indicates significantly elevated fall risk; MCID = 4 points. Why I choose it: The BBS is the gold-standard functional balance measure for stroke rehabilitation in the community. With two near-miss falls in the last month, documenting balance severity with a validated, widely-used tool is essential for justifying ongoing physiotherapy, communicating risk to the GP and MDT, and setting measurable rehabilitation goals. Critically, the BBS items include tandem standing, turning 360°, and reaching — the exact functional challenges Mr. Ibrahim reports — making it directly ecologically relevant to his presentation. How to perform:

Test 2 — Timed Up and Go Test (TUG)

What it measures: Functional mobility — time to rise from a chair, walk 3 metres, turn, return and sit. Normative values for healthy adults aged 60+ ≈ 9–10 seconds; >13.5 seconds = significantly elevated fall risk. MCID = 2.9 seconds post-stroke. Why I choose it: The TUG specifically captures the transitions Mr. Ibrahim reports as his highest-risk moments: rising from a chair, turning, and walking. It is quick, practical, and can be repeated at every session to track progress. I will also administer the Dual-Task TUG (walking + counting backwards from 20 by 3s) to simulate real-world cognitive-motor demands — in cerebellar ataxia, dual-task performance degrades significantly because the damaged integration system has reduced spare capacity, and this directly reflects community mobility ability. How to perform:

Outcome Measures Summary

Measure Purpose Timing

■ TREATMENT PLAN

- "Does the unsteadiness get worse with your eyes closed, in the dark, or in busy places like a supermarket or shopping centre?"

• "Have you had any actual falls? Any injuries? And has any of this made you cut back on activities you used to enjoy?"
• "What activities or goals matter most to you to get back? What does a good day look like for you?"
• "Have you noticed any new symptoms — new weakness, changes to your vision, headaches, or anything different from what you've been experiencing?"
• Baseline and 6 weeks
• Core Neurophysio Interventions:
• Task-specific cerebellar coordination training (adapted Frenkel's exercises):
• Highly repetitive, progressively demanding coordination tasks targeting the specific impairments — dysmetria and trunk ataxia.
• Progressive balance retraining using sensory manipulation (mCTSIB-guided):
• Begin at the condition where the patient is just at the limit of stability; systematically progress by removing vision, challenging the surface, and adding conflicting visual backgrounds.
• Gait retraining — systematic base-of-support narrowing with directional challenge:
• Progressively narrow walking stance, add deliberate turns (beginning at 45°, progressing to 90° and 180°), change surfaces (carpet, grass, gravel), add obstacles.